Research Statement

My research program lies at the intersection of algorithms, machine learning, bioinformatics and computational biology. Broadly, my work focuses on developing mathematically grounded, computationally efficient and scalable methods for extracting biological insights from increasingly large and complex molecular sequencing and biomedical datasets. Such methods are often implemented into practical tools, and sometimes even reproducible data analysis pipelines, for biomedical discovery and public health decision-making. Over the course of my career, I have worked on a range of foundational and applied problems in bioinformatics, including genome assembly, haplotype reconstruction, tumor phylogenetics, sequence representation learning, and viral epidemiology. More recently, my research has expanded toward the integration of deep learning and language-model-based techniques to better understand the mechanisms behind the evolution of different viruses and cancer types.

Overall, my publication record demonstrates sustained productivity and strong collaborative engagement. To date, I have published over 100 peer-reviewed articles, with more than 2500 citations and an h-index of 23 (cf. Google Scholar). My work appears in venues such as Bioinformatics, Journal of Computational Biology, IEEE/ACM conferences, Neural Computing and Applications, Machine Learning, Knowledge-Based Systems, Information Sciences, PAKDD, ICONIP, ISBRA, etc. Importantly, many of these publications include graduate and undergraduate student co-authors, reflecting my strong commitment to integrating research with mentorship and student success. While I have secured some external funding for my research, I expect to see more with the recent interest in artificial intelligence techniques among funding agencies. Moreover, the study of viral epidemics, driven by human behavior, will always be worthwhile despite (or maybe in spite of) the policies (or lack thereof) in place for addressing them.

Background. It was in graduate studies where I began to think much more deeply and critically about computation, pushing the envelope on what is known, publishing my first peer-reviewed research paper titled *"Grounding for model expansion in k-guarded formulas with inductive definitions"* [1] in IJCAI. Graduate studies provided the rigorous theoretical foundation that continues to inform my research to date. During my period as a postdoc, I folded in the programming skills from my undergrad studies to push my research into the more applied field of bioinformatics, with the design of practical algorithms, usable tools, and reproducible data analysis pipelines, publishing *"WhatsHap: Haplotype assembly for future-generation sequencing reads"* [2] in RECOMB, the flagship bioinformatics methods conference.

Now I operate a research lab of four PhD students and several MSc/undergrad students, as an assistant professor at Georgia State University. Here, my research has taken on big data and artificial intelligence flavors, as more bioinformatics data is produced, and more technologies for processing such data develop. It is especially important to have a solid foundation of problem solving and computational thinking skills in order to stay grounded and to quickly adapt in this rapidly evolving field. With my group we: study the diversity and dynamics of viruses from sequencing and health informatics data; devise evolutionary models for cancer progression; and computationally design antibodies for target antigens. I outline some of the main themes of this research in the following.

Representation Learning. Since starting my role as tenure-track faculty, my research program began to increasingly focus on the design of numerical feature-vector representations of molecular sequences for downstream machine learning tasks such as clustering and classification, starting with the first paper I published as a supervisor of a PhD student, Sarwan Ali, titled *"Spike2Vec: An efficient and scalable embedding approach for COVID-19 spike sequences"* [3] in IEEE BigData. This direction emerged from the practical challenge that traditional approaches struggle to scale effectively to modern sequencing datasets, especially during rapidly evolving public health situations such as the COVID-19 pandemic. Here, the goal is to design a representation that captures the important features of the sequence in a compact way, to allow downstream analysis to be efficient and scalable, but also meaningful. Since this paper, my group and I have developed dozens of embedding and representation-learning approaches for viral and protein

sequence analysis. These approaches combine ideas from kernel methods, geometry, graph methods, and natural language processing, to produce compact yet informative representations of molecular sequences. Later in this direction, we began to apply more deep-learning based techniques, to tease even more performance out of classification and clustering, with papers such as: *"Spike2CGR: an efficient method for spike sequence classification using chaos game representation"* [4] in Machine Learning (Journal). Our methods have demonstrated strong performance in tasks including viral host prediction, SARS-CoV-2 variant and protein classification, DNA-protein binding prediction, and cancer-related sequence analysis.

Benchmarking Robustness. Another important and related theme of my recent work has been robustness and reliability in machine learning for biological data. During the COVID-19 pandemic, my collaborators and I investigated the robustness of machine learning classifiers on SARS-CoV-2 genome sequences under conditions such as sequencing errors and perturbations. In this, we produced a handful of papers, including *"Benchmarking machine learning robustness in Covid-19 genome sequence classification"* [5], published in Scienfic Reports. These studies highlighted important limitations in current machine learning approaches while also proposing methods for improving resilience and generalization.

Tumor Phylogeny. My research in cancer genomics and tumor phylogeny focuses on developing computational and machine learning methods to reconstruct the evolutionary history of tumors from high-throughput and single-cell sequencing data. This work addresses key challenges such as clustering noisy single-cell data, modeling mutation accumulation and loss, simulating tumor evolution, and benchmarking phylogeny reconstruction pipelines. A notable paper in this area is *"Effective clustering for single cell sequencing cancer data"* [6], published in the IEEE Journal of Biomedical and Health Informatics. By combining algorithm design, graph-based methods, and modern representation learning techniques, this research aims to improve our understanding of cancer progression, metastasis, and intratumor heterogeneity, ultimately supporting more accurate precision oncology and cancer diagnostics.

Antibody Design. More recently, with PhD student Mansoor Ahmed, we work on computational antibody design, which combines machine learning, protein language models, and biological domain knowledge to improve the prediction and optimization of antibody-antigen interactions. His research includes the development of benchmark datasets, graph-based learning methods, and deep generative models for epitope prediction and antibody design, with applications in infectious disease and therapeutic discovery and design. By leveraging modern AI techniques together with biological sequence analysis, his work aims to accelerate the discovery of effective and robust candidate antibodies for a target antigen while improving the interpretability and evaluation of computational antibody engineering methods. A representative paper in this area is our recent work titled *"Improved graph-based antibody-aware epitope prediction with protein language model-based embeddings"* [7], published in the postproceedings of ICCABS 2025. We had a very recent presentation of the latest from this area of work in the GEM 2026 ICLR Workshop.

Multidisciplinary Collaboration. A defining feature of my research program is its multidisciplinary nature. My collaborations span computer science, biology, epidemiology, medicine, and public health. I have worked with researchers studying viral genomics, cancer evolution, protein function, transcriptomics, acoustics, and epidemiological modeling. These collaborations have broadened the impact of my work and created opportunities for students to engage in multidisciplinary research environments. At GSU, I have collaborated with faculty across departments and institutes, including work connected to HIV transmission analysis, machine learning robustness, molecular sequence analysis, and graph-based biological data analytics. I have also maintained international collaborations developed during my postdoctoral appointments in Europe. These collaborations continue to contribute to joint publications and proposals.

My research program also emphasizes methodological transfer across domains. Many of the representation-learning and embedding techniques developed in my group have applications beyond biology, including graph analytics, privacy-preserving federated learning, natural language processing, and health informatics. I view this cross-disciplinary adaptability as one of the strengths of the research program. An

example of this is a recent such paper which has nothing to do with bioinformatics: *"Bounded conversational memory with hybrid retrieval and evidence highlighting for multi-session dialogue systems"* [8], published recently to SIGDIAL. Yet, ideas from this language modeling perspective inform my research in bioinformatics, in particular language model embedding-based representations of molecular sequences, which is a powerful concept for capturing meaningful features of the sequences.

Student Success. An important component of my scholarly activity is mentoring students at all levels. To date, I have supervised numerous PhD students, Master's students, undergraduate researchers, and honors thesis students. Several of my students have received fellowships, dissertation awards, research awards, and competitive recognitions. Multiple students have successfully defended theses and dissertations under my supervision, and many have co-authored publications in peer-reviewed journals and conferences. I place significant emphasis on involving students directly in active research projects. Students in my group participate in all stages of the research process, including literature review, algorithm development, software implementation, experimentation, manuscript preparation, and conference submission. This mentorship philosophy is designed not only to produce strong research outcomes, but also to prepare students for careers in academia and industry. I am particularly committed to broadening participation in artificial intelligence and computational biology research. My participation in initiatives such as the AI Learning Alliance and NSF-supported REU activities reflects my interest in expanding access to research opportunities for students from historically underrepresented backgrounds. I believe that creating accessible and collaborative research environments is essential for advancing both scientific innovation and student success.

Future Goals

Some of my future goals during the next five years to extend my current research are as follows.

Evolutionary Modeling from Sequencing Big Data. A central goal of my future work is to develop new computational models for understanding non-classical patterns of evolution observed in large-scale sequencing data. Modern sequencing datasets reveal evolutionary phenomena that are not adequately captured by traditional phylogenetic assumptions, including reversions, saltational events, convergent evolution, and complex epistatic interactions. I plan to develop methods that integrate phylogenetics, population genetics, probabilistic modeling, and deep learning to better characterize these processes. This work will include the development of scalable algorithms for identifying unusual evolutionary trajectories in viral and tumor data, with applications in infectious disease surveillance and precision oncology.

AI and Language Models for Biological Sequences. Another major direction will involve the application of foundation models and language-model-inspired architectures to biological sequence analysis. Biological sequences possess many structural similarities to natural language, and I believe that large-scale representation learning methods will fundamentally reshape computational biology. My group will investigate methods that combine biological priors with transformer-based and graph-based architectures for tasks such as protein classification, epitope prediction, host specification, mutation impact prediction, and evolutionary forecasting. I am especially interested in interpretable and biologically grounded AI methods that can provide mechanistic insight rather than solely predictive performance.

Public Health and Epidemiological Applications. I also plan to continue developing computational methods for public health applications, particularly involving viral transmission dynamics and epidemiological surveillance. This includes integrating genomic data with social and behavioral information to better understand pathogen spread and evolutionary adaptation. Recent proposal efforts examining semantic shifts in human-generated text alongside genomic evolution represent an exciting new interdisciplinary direction. Over the next five years, I aim to establish a broader research framework connecting machine learning, genomics, epidemiology, and social behavior analysis.

Conclusion. My research program combines algorithmic foundations and machine learning to address important bioinformatics problems in genomics and evolution. Through multidisciplinary collaboration, sustained publication activity, external funding efforts, and strong student mentorship, I have established a productive and growing research agenda. Over the next five years, I intend to continue advancing computational methods for biological sequence analysis and evolutionary modeling—something which will need to be continuously fine-tuned as more epidemics arise, along with the resulting data collected. I also plan to continue expanding opportunities for student involvement, with various university-level, locally-sponsored, and national programs. I believe these efforts will contribute both to scientific discovery and to the development of the next generation of bioinformatics researchers.

References

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